

Closed-Loop Thermodynamic Stability Monitoring Provides Early Warning Before Bearing Failure in Run-to-Failure Trajectories

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May 2026

Abstract

We report that continuous monitoring of a thermodynamic stability metric, $\Phi = I \times \rho - \alpha \times S$, detects bearing failure risk substantially before end of life in all 10 tested XJTU-SY run-to-failure trajectories. With a fixed coupling constant $\alpha = 0.1$ and critical threshold $\Phi_c = 0.25$, the metric crossed the critical threshold at an average of 86.1 percent of bearing lifetime remaining, with a minimum of 73.2 percent and a maximum of 90.4 percent, across lifecycles spanning 42 to 2,538 measurement files. No per-bearing training, no failure history, and no threshold adjustment were used. We also report two narrower extensions. First, the same formula and threshold correctly classified a cross-domain subset of 11 systems across mechanical, aerospace, and geophysical domains with no threshold adjustment between domains. Second, we derive and validate an inverse design constraint from the stability condition on the same 11 systems, with the caveat that only one stable case was available for the inverse design validation. All experiments use real published datasets with no synthetic data. Negative results and boundary conditions are reported throughout.

1 Introduction

Bearing failure causes unplanned industrial downtime and safety incidents. The practical question for maintenance engineers is not whether a bearing will eventually fail but how early a monitoring system can reliably signal elevated risk. A warning that arrives with most of bearing life remaining allows scheduled replacement. A warning that arrives in the final few percent of life forces emergency response.

Existing bearing prognostics methods fall into two categories. Data-driven methods train regression models on historical failure data to estimate remaining useful life. Physics-based methods track vibration spectral features against domain-specific thresholds. Both require either failure history from the specific bearing population or domain expertise to interpret spectral signatures. Neither produces a domain-agnostic scalar stability metric that can be computed from raw vibration data without per-bearing calibration.

This paper asks whether a thermodynamic stability metric computed from three quantities (information preservation, temporal coherence, and entropy) can detect bearing failure risk early enough to be operationally useful, without any training on bearing failure data.

The metric is $\Phi = I \times \rho - \alpha \times S$, where I is the ratio of baseline to current vibration RMS, ρ is the temporal autocorrelation of the vibration signal, S is the Shannon entropy of the measurement

distribution, and $\alpha = 0.1$ is a fixed coupling constant. The critical threshold is $\Phi_c = 0.25$. These constants were carried over unchanged from earlier thermodynamic stability validation work and were not re-fitted within the closed-loop bearing analysis reported here.

The central finding is that across all 10 tested XJTU-SY bearings, the metric dropped below the critical threshold at an average of 86.1 percent of total bearing lifetime remaining, with a minimum advance warning of 73.2 percent and a maximum of 90.4 percent. The result holds across bearing lifecycles spanning more than one order of magnitude, from 42 to 2,538 measurement files.

We also report two extensions. First, the same formula and threshold correctly classified 11 systems across mechanical, aerospace, and geophysical domains in a cross-domain platform test, with no threshold adjustment between domains. Second, we derive an inverse design constraint from the stability condition that correctly partitioned stable and failing systems in the tested set, with the caveat that only one stable case was available for the inverse design validation.

The paper makes the following contributions:

- We show that $\Phi = I \times \rho - \alpha \times S$ with fixed constants drops below threshold at 73.2 to 90.4 percent of bearing lifetime remaining across all 10 tested XJTU-SY run-to-failure trajectories, with no per-bearing calibration.
- We show that the same formula and threshold correctly classified 11 systems across mechanical, aerospace, and geophysical domains in a cross-domain platform test.
- We derive and validate an inverse design constraint $I \times \rho > 0.25 + \alpha \times S$ on 11 systems across three domains, with documented limitations on the stable-case sample size.
- We document boundary conditions and negative results throughout, including the shortest-lifecycle bearing producing the minimum advance warning and the single stable case available for inverse design validation.

All experiments use real published datasets. No synthetic data was used at any point. Negative results are reported throughout.

2 The Thermodynamic Stability Metric

2.1 Definition

The thermodynamic stability metric Φ is computed from three quantities derived from sensor measurements:

$$\Phi = I \times \rho - \alpha \times S \quad (1)$$

where I is information preservation, ρ is temporal coherence, S is Shannon entropy in bits, and $\alpha = 0.1$ is a fixed coupling constant.

Information preservation I quantifies how well the system maintains its baseline operational signature:

$$I = \frac{\text{RMS}_{\text{baseline}}}{\text{RMS}_{\text{current}}} \quad (2)$$

where RMS is the root mean square of the vibration signal. As a bearing degrades, vibration amplitude increases, driving $\text{RMS}_{\text{current}}$ above the baseline and reducing I toward zero. I is capped at 1.0 for cases where current RMS falls below baseline.

Temporal coherence ρ is the Pearson autocorrelation of the vibration signal at lag 1:

$$\rho = \text{corr}(x_t, x_{t+1}) \quad (3)$$

High autocorrelation indicates smooth, predictable signal evolution characteristic of healthy operation. As degradation progresses, the signal becomes more erratic and ρ decreases.

Entropy S is the Shannon entropy of the measurement distribution in bits, computed by discretizing the vibration signal into histogram bins and applying:

$$S = - \sum_i p_i \log_2 p_i \quad (4)$$

where p_i is the fraction of measurements in bin i . Higher entropy indicates a more dispersed, unpredictable signal.

2.2 Stability Classification

The critical threshold $\Phi_c \approx 0.25$ separates stable from unstable states. Systems with $\Phi \geq 0.25$ are classified as stable. Systems with $0 \leq \Phi < 0.25$ are in a critical state with elevated failure risk. Systems with $\Phi < 0$ are in a severe instability state. The threshold was identified empirically in prior validation work across multiple domains and was carried over unchanged for the bearing analysis reported here.

2.3 What the Metric Measures and What It Does Not

Φ measures the approach to failure as a continuous degradation signal. It does not estimate remaining useful life as a quantity in time or measurement cycles. It does not classify between two stable operating modes. This boundary condition is relevant for interpreting results: Φ is best treated as a transition detector rather than as a general classifier of multiple stable operating modes.

3 Datasets and Experimental Protocol

3.1 Dataset

All experiments use the XJTU-SY bearing dataset (Wang et al., 2018), a publicly available run-to-failure vibration benchmark from Xi'an Jiaotong University. The dataset contains bearings operated under three controlled conditions until failure, with vibration measurements recorded at regular intervals throughout each bearing's operational lifetime.

Ten bearings were analyzed across three operating conditions: Condition 1 at 35 Hz and 12 kN radial load (Bearings 1_1 through 1_4), Condition 2 at 37.5 Hz and 11 kN radial load (Bearings 2_2 through 2_5), and Condition 3 at 40 Hz and 10 kN radial load (Bearings 3_1 and 3_4). Bearing lifecycles in the tested set ranged from 42 to 2,538 measurement files, spanning more than one order of magnitude.

3.2 Experimental Protocol

For each bearing, the following steps were executed in order:

1. Establish a baseline RMS from the first measurement file in the bearing's recorded lifecycle.

2. At each subsequent timestep, compute I , ρ , and S from the current measurement window.
3. Compute Φ using Equation 1.
4. Record the first timestep at which Φ drops below the warning threshold (0.30) and the critical threshold (0.25).
5. Compute advance warning as the fraction of total lifecycle remaining after the first threshold crossing.

No failure labels from the bearing were used at any point. The coupling constant $\alpha = 0.1$ and threshold $\Phi_c = 0.25$ were fixed before analysis and not adjusted for any individual bearing. All computations were implemented in Python using NumPy for signal processing and SciPy for entropy calculation.

3.3 Advance Warning Calculation

Advance warning is reported as a percentage of total bearing lifetime:

$$\text{Advance Warning} = \frac{T_{\text{total}} - T_{\text{alert}}}{T_{\text{total}}} \times 100\% \quad (5)$$

where T_{total} is the total number of measurement files in the bearing’s lifecycle and T_{alert} is the timestep at which Φ first crossed the threshold. A higher percentage means the alert fired earlier in the bearing’s life.

4 Closed-Loop Advance Warning Results

4.1 Per-Bearing Results

Table 1 shows the advance warning percentage for each of the 10 tested bearings at both the warning threshold ($\Phi < 0.30$) and the critical threshold ($\Phi < 0.25$).

All 10 bearings crossed both the warning and critical thresholds before end of life. Critical advance warning ranged from 73.2 percent (Bearing2_4, 42-file lifecycle) to 90.4 percent (Bearing1_3, 158-file lifecycle), with an average of 86.1 percent. The warning threshold ($\Phi < 0.30$) produced slightly earlier detection on average (86.6 percent) with a minimum of 75.6 percent.

4.2 Consistency Across the Tested Lifecycle Range

The advance warning percentages are consistent across bearing lifecycles spanning more than one order of magnitude. Bearing2_4 with 42 measurement files and Bearing3_1 with 2,538 measurement files both produced advance warnings in the same range (73.2 percent and 90.0 percent respectively), showing that the effect is not confined to one lifecycle length in the tested set.

4.3 Boundary Conditions

Bearing2_4 produced the minimum advance warning at 73.2 percent and has the shortest lifecycle in the dataset at 42 measurement files. Both the shortest lifecycle and the minimum advance warning come from the same bearing, which is a potential confound. Whether the method provides equivalent advance warning on bearings with very short degradation trajectories under different operating conditions is not established by this dataset.

Bearing	Lifecycle (files)	Final Φ	Warning Advance ($\Phi < 0.30$)	Critical Advance ($\Phi < 0.25$)
Bearing1_1	123	-0.178	82.0%	82.0%
Bearing1_2	161	-0.335	90.0%	90.0%
Bearing1_3	158	-0.295	90.4%	90.4%
Bearing1_4	122	-0.005	83.5%	83.5%
Bearing2_2	161	-0.268	90.0%	90.0%
Bearing2_3	533	-0.381	90.0%	90.0%
Bearing2_4	42	-0.255	75.6%	73.2%
Bearing2_5	339	-0.371	84.6%	82.2%
Bearing3_1	2538	-0.275	90.0%	90.0%
Bearing3_4	1515	-0.288	90.0%	90.0%
Average			86.6%	86.1%
Minimum			75.6%	73.2%

Table 1: Per-bearing advance warning results across 10 XJTU-SY run-to-failure trajectories. Warning advance is the percentage of bearing lifetime remaining when Φ first dropped below 0.30. Critical advance is the percentage remaining when Φ first dropped below 0.25. All 10 bearings were detected at both thresholds before end of life. Final Φ values are from the closed-loop monitoring script.

Bearing1_4 produced a final Φ of -0.005, the closest to zero of any bearing in the tested set. Its critical advance warning of 83.5 percent is within the normal range, but the near-zero final Φ means this bearing ended near the boundary between the critical and collapse regimes.

5 Component Analysis

Table 2 shows the end-of-life values of the three Φ components for each bearing, drawn from the reference validation implementation. Final Φ values for each bearing are reported in Table 1.

Bearing	I	ρ	S (bits)	Operating Condition
Bearing1_1	0.086	0.978	2.522	35 Hz, 12 kN
Bearing1_2	0.052	0.992	3.875	35 Hz, 12 kN
Bearing1_3	0.072	0.989	3.323	35 Hz, 12 kN
Bearing1_4	0.248	0.505	1.281	35 Hz, 12 kN
Bearing2_2	0.066	0.994	3.281	37.5 Hz, 11 kN
Bearing2_3	0.109	0.997	4.782	37.5 Hz, 11 kN
Bearing2_4	0.058	0.988	3.122	37.5 Hz, 11 kN
Bearing2_5	0.095	0.996	4.393	37.5 Hz, 11 kN
Bearing3_1	0.043	0.992	3.190	40 Hz, 10 kN
Bearing3_4	0.118	0.997	4.073	40 Hz, 10 kN

Table 2: End-of-life Φ component values for all 10 tested bearings from the reference validation implementation. I = information preservation (baseline RMS / final RMS). ρ = temporal auto-correlation at lag 1. S = Shannon entropy in bits. Final Φ values are reported in Table 1.

In 9 of 10 bearings, the drop in Φ at end of life is driven primarily by the reduction of I toward zero while ρ remains high (at or above 0.978 in 9 cases). This indicates that temporal coherence is maintained even as vibration amplitude diverges from baseline, meaning the signal remains structured even as it departs from healthy behavior.

Bearing1.4 is the exception. Its end-of-life ρ of 0.505 is substantially lower than all other bearings, and its I of 0.248 is the highest of the group. This bearing’s failure is characterized by loss of temporal coherence rather than amplitude reduction, producing a qualitatively different degradation signature while still resulting in a sub-threshold Φ .

6 Cross-Domain Platform Validation

The bearing advance warning result is the primary finding of this paper. We report one extension to test whether the same formula and threshold produce outcome-consistent classifications across other physical domains without modification.

6.1 Setup

Eleven systems were evaluated across three domains: four XJTU-SY bearings (Bearings 1_1, 1_2, 2_3, and 3_1), four NASA C-MAPSS turbofan engines (Saxena et al., 2008) (Engines 1 through 4), and three geophysical cases from the USGS Donna Lea strainmeter (USGS, 2016) (Parkfield M6.0 2004, San Simeon M6.5 2003, and the 2010 quiet seismic year). The same formula, the same $\alpha = 0.1$, and the same threshold $\Phi_c = 0.25$ were applied to all 11 systems with no domain-specific adjustment.

6.2 Results

Table 3 summarizes the cross-domain platform results.

System	Domain	Φ	Prediction	Outcome
Bearing1.1	Mechanical	-0.168	COLLAPSE	FAILURE
Bearing1.2	Mechanical	-0.336	COLLAPSE	FAILURE
Bearing2.3	Mechanical	-0.370	COLLAPSE	FAILURE
Bearing3.1	Mechanical	-0.276	COLLAPSE	FAILURE
Engine_1	Aerospace	0.205	CRITICAL	FAILURE
Engine_2	Aerospace	0.186	CRITICAL	FAILURE
Engine_3	Aerospace	0.152	CRITICAL	FAILURE
Engine_4	Aerospace	0.241	CRITICAL	FAILURE
Parkfield M6.0	Geophysical	0.114	CRITICAL	FAILURE
San Simeon M6.5	Geophysical	0.084	CRITICAL	FAILURE
Stable 2010	Geophysical	0.577	STABLE	STABLE
Overall			11/11	100%

Table 3: Cross-domain platform validation results across 11 systems in three domains. The same Φ formula, $\alpha = 0.1$, and threshold $\Phi_c = 0.25$ were applied to all systems with no domain-specific adjustment. The 2010 quiet seismic year provides the one stable case in this validation subset.

All 11 systems were classified consistently with their observed outcomes. The four bearings fall

in the collapse regime ($\Phi < 0$), the four turbofans and two earthquake events fall in the critical regime ($0 < \Phi < 0.25$), and the 2010 quiet seismic year falls in the stable regime ($\Phi = 0.577$).

6.3 Scope

This cross-domain result uses 11 systems drawn from the same validation sets used in prior thermodynamic stability work. It is not an independent held-out dataset. The result establishes that the same formula and threshold produce outcome-consistent classifications across three physical domains in the tested subset, not that the formula has been independently validated on new systems outside the prior validation history.

7 Inverse Design Constraint

7.1 Derivation

The stability condition $\Phi > 0.25$ can be rearranged algebraically to produce a design-time constraint. Substituting the definition of Φ :

$$I \times \rho - \alpha \times S > 0.25 \tag{6}$$

Rearranging:

$$I \times \rho > 0.25 + \alpha \times S \tag{7}$$

A system satisfying this constraint has an information-coherence product that exceeds the stability requirement at the measured entropy level. In the tested set, systems violating the constraint fell below the stability threshold.

7.2 Results

Table 4 shows the inverse design constraint evaluation for the same 11 systems.

System	Domain	$I \times \rho$	Required	Constraint
Bearing1_1	Mechanical	0.084	0.502	VIOLATES
Bearing1_2	Mechanical	0.052	0.638	VIOLATES
Bearing2_3	Mechanical	0.108	0.728	VIOLATES
Bearing3_1	Mechanical	0.043	0.569	VIOLATES
Engine_1	Aerospace	0.439	0.485	VIOLATES
Engine_2	Aerospace	0.435	0.499	VIOLATES
Engine_3	Aerospace	0.397	0.496	VIOLATES
Engine_4	Aerospace	0.476	0.484	VIOLATES
Parkfield M6.0	Geophysical	0.434	0.571	VIOLATES
San Simeon M6.5	Geophysical	0.415	0.581	VIOLATES
Stable 2010	Geophysical	0.881	0.555	MEETS
Overall				11/11

Table 4: Inverse design constraint evaluation for all 11 systems. Required $I \times \rho$ is computed as $0.25 + 0.1 \times S$ for each system. Systems where actual $I \times \rho$ exceeds the required value are classified MEETS (stable). All others are classified VIOLATES (failure). 10/10 failed systems violated the constraint. The one stable system met it.

All 10 failed systems violated the constraint and the one stable system met it, producing 11/11 outcome-consistent classifications. The violation margin varied substantially across systems, from Engine_4 at 0.008 (narrowest, 0.476 vs required 0.484) to Bearing1_2 at 0.586 (widest, 0.052 vs required 0.638).

7.3 Scope and Limitations

The inverse design validation has one stable case. A false positive rate cannot be estimated from one stable system. All aerospace engines tested were near end of life, so no mid-life aerospace systems were available to test whether healthy systems correctly satisfy the constraint. The inverse design test uses the same data as the cross-domain platform test, meaning the two results are not independent. This section reports an empirically validated algebraic constraint, not a theorem-level stability proof.

8 Related Work

Bearing prognostics. Data-driven remaining useful life estimation for rotating machinery is a well-studied problem. Lei et al. (Lei et al., 2018) survey machine learning approaches for machinery health monitoring, covering vibration feature extraction and degradation modeling. Nectoux et al. (Nectoux et al., 2012) introduce the PRONOSTIA benchmark for bearing prognostics. These methods typically require failure-labeled training data from the specific bearing population being monitored. The method reported here does not use failure-labeled training data from the monitored bearing population and does not estimate remaining useful life as a point prediction.

Physics-informed and model-based prognostics. Jardine et al. (Jardine et al., 2006) review condition monitoring and diagnostics approaches based on vibration analysis and physical models. Sikorska et al. (Sikorska et al., 2011) compare prognostic models including physics-based and data-driven approaches. These methods require domain expertise and system-specific param-

terization. The thermodynamic stability metric used here is computed from raw vibration statistics without physical modeling of bearing dynamics.

Thermodynamic and entropy-based approaches. Amiri and Modarres (Amiri and Modarres, 2014) apply entropy generation to fatigue damage assessment in metals, measuring cumulative thermodynamic entropy during cyclic loading. Yun et al. (Yun et al., 2019) apply thermodynamic concepts to bearing degradation assessment. These approaches are domain-specific and require physical temperature measurements or material-specific thermodynamic equations. The metric reported here uses Shannon entropy of the measurement distribution rather than thermodynamic entropy of the physical process and does not require temperature instrumentation.

Cross-domain stability assessment. The cross-domain extension reported in Section 6 builds on prior work validating the same Φ formula across mechanical, electrical, aerospace, and geophysical systems (Barnicle, 2026). The present paper focuses on the closed-loop bearing advance warning result, which was not the primary focus of that prior work.

9 Discussion and Limitations

9.1 What This Work Shows

The central finding is that $\Phi = I \times \rho - \alpha \times S$ with fixed constants drops below the critical threshold at an average of 86.1 percent of bearing lifetime remaining across all 10 tested XJTU-SY run-to-failure trajectories, with a minimum of 73.2 percent. No per-bearing training, no failure history, and no threshold adjustment were used.

The component analysis shows that in 9 of 10 bearings, the drop in Φ is driven primarily by reduction of I toward zero while ρ remains high, indicating that temporal coherence is maintained as vibration amplitude diverges from baseline. Bearing1_4 is the exception, showing loss of temporal coherence as the dominant failure mechanism.

The cross-domain extension and inverse design results are narrower claims. Both use the same 11 systems from prior validation work and are not independent validations.

9.2 Limitations

Single dataset. All bearing results come from the XJTU-SY dataset under three controlled operating conditions. Whether the advance warning percentages generalize to other bearing types, other failure modes, or other operating conditions is not established by this paper.

Baseline selection. The baseline RMS is established from the first measurement file in each bearing’s recorded lifecycle. Different baseline choices would produce different Φ trajectories and different advance warning percentages. The sensitivity of results to baseline selection has not been tested.

Bearing2_4 confound. The shortest lifecycle bearing (42 files) produced the minimum advance warning (73.2 percent). Both the minimum lifecycle and minimum advance warning come from the same bearing, making it difficult to determine whether short lifecycles systematically produce shorter advance warnings.

Fixed thresholds. The warning threshold (0.30) and critical threshold (0.25) were carried over from prior work and were not optimized for bearing detection specifically. Different threshold choices would produce different advance warning percentages. The sensitivity of results to threshold selection has not been tested.

Inverse design one stable case. The inverse design validation has one stable system. A false positive rate for the design constraint cannot be estimated from one case.

Cross-domain non-independence. The cross-domain platform and inverse design results share the same 11 systems. They are not independent validations of the same claim.

9.3 Future Work

The most direct extension is to evaluate the closed-loop advance warning result on additional bearing datasets with different failure modes, operating conditions, and lifecycle lengths, to determine whether the 73.2 to 90.4 percent advance warning range holds outside the XJTU-SY dataset.

A second direction is to test the sensitivity of advance warning percentages to baseline selection strategy, including rolling baselines and multi-file initialization windows, to understand how much the result depends on the specific baseline protocol used here.

A third direction is to include stable bearing cases, meaning bearings that were monitored but did not fail during the observation period, to evaluate the false positive rate of the method under continuous monitoring conditions.

Finally, the cross-domain extension would benefit from independent held-out system sets not drawn from prior validation history, and the inverse design validation would benefit from a larger stable-case sample across multiple domains.

Code Availability

The validation scripts, raw output logs, and result files for the closed-loop bearing analysis are available at <https://github.com/Wise314/universal-stability-engineering>.

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Patent Disclosures

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